

When Does the LLM Disambiguation Stage Actually Help? A Decomposition of Retriever–LLM Interaction in Biomedical Entity Linking

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Abstract

Large language models are routinely bolted onto biomedical entity-linking (EL) pipelines as a final disambiguation stage, and recent work reports large gains from doing so. We do not propose a new method: we ask *how much*, *when*, and *why* this stage helps. Using a re-produced neuro-symbolic pipeline as a measuring instrument, we run one fixed open 4B disambiguator behind fourteen retriever configurations spanning five corpora (BC5CDR, BioRED, NCBI-Disease, NLM-Chem, Med-Mentions) and four retrievers, including two public systems we did not build. Three results follow. (i) The stage’s contribution is set by the retriever, not by the model: it ranges from +0.6 to +10.1 Accuracy@1 points with the same model and prompt, and *within* a corpus a stronger retriever reliably means a smaller gain (12/14 ordered pairs; dataset-demeaned $\rho = -0.72$, $p = 0.004$). Degrading one retriever’s ranking in graded steps on a fixed candidate pool makes this causal: the stage absorbs 89% of the accuracy the retriever gives up, linearly ($r = -1.0000$). (ii) The zero-shot model’s intervention rate is well calibrated but its intervention precision is not: precision collapses in a *mid-confidence* band where the retriever is already right, making the stage net-harmful there; this survives three prompt ablations, including one that states the retriever’s margin as a number and instructs the model to defer on large margins. (iii) LoRA fine-tuning does not improve the model where disambiguation is hard—precision on the truly-uncertain band is unchanged—but installs a confidence-dependent *restraint* whose reliability benefit is bound to the training distribution: on held-out corpora it raises intervention precision from 69% to 96% on concepts seen in fine-tuning and from 72% to 74% on unseen ones. We release all prediction dumps and analysis code.

1 Introduction

Biomedical entity linking maps a mention in text to a concept in a knowledge base such as MeSH or UMLS (Bodenreider, 2004). Most systems are two-stage: a retriever proposes candidate concepts, and a disambiguation stage selects among them. Large language models (LLMs) are now the default choice for the second stage. Ye and Mitchell (2025) report a new state of the art—up to +16 points—from a training-free LLM disambiguator placed on top of an existing retriever, and Lin et al. (2025) obtain competitive results by constraining the decoder. The field treats this stage as a semantic reasoner that supplies what the retriever cannot.

A single accuracy number cannot support that reading. It hides where the stage helps, where it hurts, and what fine-tuning changes about it. It also hides how much of the reported gain is a property of the LLM at all, as opposed to a property of the retriever it was placed behind: the same disambiguator behind a weak retriever has far more to fix than behind a strong one, so “+16 points” is a statement about a pair of components, not about one of them.

We therefore reframe the problem from “build a better system” to “characterise a stage the field has not yet characterised”. We reproduce the neuro-symbolic BioLinkerAI pipeline (Sakor et al., 2025) and use it as an instrument: we hold the disambiguator, the prompt and the mention set fixed and vary only the retriever, across fourteen configurations and five corpora. Our contributions are:

1. **The frame (§4).** The stage’s marginal value is a decreasing function of retriever strength. Ye and Mitchell (2025) note this qualitatively over two base models; we measure it over fourteen configurations, show it is identified *within* a corpus and not across corpora, and then manipulate it directly: degrading one retriever’s ranking in graded steps on a fixed

candidate pool, the stage absorbs 89% of the accuracy lost, linearly ($r = -1.0000$).

2. **A mid-confidence danger zone (§5).** The intervention *rate* falls with retriever confidence but the intervention *precision* collapses in a middle band, where the model over-rides a mostly-correct retriever. It survives three prompt ablations, a surface-similarity control and an external retriever’s confidence signal.
3. **Deference, not competence (§6).** Fine-tuning leaves precision flat on the cases the retriever cannot solve and instead learns *when* not to act; the reliability it buys is realised almost entirely on concepts present in the fine-tuning data.

2 Related Work

Biomedical EL and the LLM stage. BioLinkerAI (Sakor et al., 2025) is a neuro-symbolic pipeline (linguistic rules, candidate generation, domain rules, LLM disambiguation) that we reproduce as our instrument. Ye and Mitchell (2025) isolate the LLM as a disambiguator on top of SapBERT and ArboEL and observe qualitatively, over two fixed base models, that its value depends on the base model’s strength; we make that relation quantitative, identify it within corpora, and extend it to fine-tuning. Lin et al. (2025) guide the LLM with restrictive and contrastive decoding; their own ablations show the LLM machinery adds ≤ 0.3 points on BC5CDR, which is indirect support for our frame. The current state of the art is a trained generative model, ANGEL (Kim et al., 2025); it has no separable retriever stage and is orthogonal to our analysis. Our retrievers are a lexical/BM25 baseline, SapBERT (Liu et al., 2021), BioSyn (Sung et al., 2020) and the reproduced rule pipeline.

Selective prediction and confidence gating. Abstaining on low-confidence inputs is long-standing in selective classification (Geifman and El-Yaniv, 2017) and, for LLM pipelines, in cascaded routing (Chen et al., 2023). Our setting inverts the polarity: we abstain from *intervening* rather than from predicting, and the confidence signal belongs to the upstream retriever, not to the model being gated. Doku (2026) call the failure mode we observe an “inversion zone”.

Memorisation under fine-tuning. That fine-tuning trades cross-domain generalisation for in-domain performance is known for EL retrievers

(Logeswaran et al., 2019). Our contribution is the mechanism for the LLM disambiguation stage, with a zero-shot control on the same mentions: what transfers is not accuracy but restraint, and only in-distribution.

3 Experimental Setup

Pipeline. We use our reproduction of BioLinkerAI: candidate generation (P2), rule-based re-ranking (P3), and LLM disambiguation (P4). P2 matches the mention against a MeSH index (descriptors and supplementary concepts) enriched with synonyms harvested from UMLS, Wikidata and DBpedia, after an abbreviation-expansion pass; P3 re-ranks with hand-written domain rules and emits a top-10 candidate list with scores. Throughout, P3 denotes the retriever’s final top-1 and P4 the pipeline’s output; the stage’s effect is $\Delta = P4 - P3$ in Accuracy@1 points. Mentions are gold-annotated: we evaluate linking, not detection. Where a run is labelled with a standalone retriever (BM25, SapBERT, BioSyn), P2 and P3 are replaced wholesale by that system’s ranked list and scores, and only P4 is ours.

Disambiguator. Qwen3-4B (Yang et al., 2025) served with vLLM (Kwon et al., 2023), temperature 0, with a fixed prompt and decoding constrained to an enumeration over the candidate list, so the stage can only select among P3’s candidates and never invents an identifier. It runs in two variants: zero-shot, and LoRA (Hu et al., 2022) supervised fine-tuning on the BC5CDR training split (chat format, completion-only loss, at most five examples per mention–concept pair, no in-context examples). All runs share the same prompt, the same decoding settings and the same top-10 candidate lists, so every difference we report is attributable to the retriever or to the fine-tuning, never to the interface.

Corpora and retrievers. BC5CDR (Li et al., 2016), BioRED (Luo et al., 2022), NCBI-Disease (Doğan et al., 2014) and NLM-Chem (Islamaj et al., 2021) are linked against MeSH; MedMentions (Mohan and Li, 2019) against the full UMLS. Following the BioLinkerAI evaluation protocol we keep duplicate mention strings, which inflates absolute accuracies relative to de-duplicated protocols (Tutubalina et al., 2020); all comparisons here are between systems on the identical mention set, so this does not affect any Δ . Table 1 lists the four-

teen retriever configurations. BM25 and BioSyn are standalone public systems we did not build; for those runs the LLM stage sees only that system’s candidates and scores. Two entries in that table need a word. On NCBI-Disease dense retrieval is *worse* than lexical matching (Recall@1 60.7 vs. 71.3): the corpus is dominated by short disease abbreviations and modifier phrases whose surface form is highly diagnostic, which SapBERT’s embedding space blurs. And the two MedMentions runs cover 4,823 and 5,000 mentions; where they are compared directly (§4) we restrict to the 4,823 they share.

A reproduction, not a replication. Our pipeline follows BioLinkerAI’s architecture but does not reach its reported accuracy: on BC5CDR we obtain 83.7 against the 93.3 reported there with GPT-4, and on MedMentions 59.6 against 81.3. Neither the code nor the prompts were released and the reported system uses a closed-source disambiguator, so we cannot attribute the gap; it may lie in candidate generation, in the knowledge-base snapshot, in the disambiguator, or in evaluation details. We therefore do not present this pipeline as a replication or as a competitive system, and we make no cross-paper accuracy claims. It is used only as a controlled instrument: every comparison we report holds the mention set, the candidate lists, the prompt and the decoder fixed and varies one component, so a constant offset in absolute accuracy does not affect any Δ . The two standalone public retrievers (BM25 and BioSyn) additionally check that the effects are not artefacts of our own candidate generation and scoring.

Confirmatory vs. descriptive analyses. The band-level breakdowns are descriptive: they slice five confidence bands across five corpora and two model variants, and we do not correct their intervals for multiplicity. Three analyses are confirmatory and carry the paper’s claims: the within-corpus relation between retriever strength and the stage’s gain (§4), the seen-versus-unseen precision contrast (§6), and the gated-versus-full comparison (§5). Each is a single pre-specified test.

Metrics. *Candidate Recall@1* is the retriever’s top-1 accuracy and our measure of retriever strength; *Recall@1@10* (the gold concept appearing anywhere in the candidate list) bounds what the stage could possibly recover. We report the *intervention rate* (share of mentions whose top-

Corpus	Retriever	<i>n</i>	R@1	R@10	P4	Δ
BC5CDR	BM25	9,391	73.4	79.1	75.4	+2.0
	SapBERT	9,661	75.9	87.3	80.6	+4.6
	pipeline	9,661	82.7	91.6	83.7	+0.9
	BioSyn	9,661	84.8	90.5	85.4	+0.6
BioRED	SapBERT	1,661	70.8	86.5	77.9	+7.1
	lexical	1,661	74.2	88.1	82.1	+7.9
	pipeline	1,661	79.5	93.6	83.6	+4.0
NCBI	SapBERT	863	60.7	78.6	64.2	+3.5
	pipeline	863	70.1	90.0	72.5	+2.4
	lexical	863	71.3	84.4	72.8	+1.5
NLM-Chem	lexical	1,500	65.9	81.0	76.0	+10.1
	dense	1,500	72.3	86.9	78.0	+5.7
MedMent.	lexical	4,823	49.6	71.4	58.4	+8.9
	SapBERT	5,000	55.3	77.3	59.6	+4.3

Table 1: The fourteen retriever configurations, ordered by retriever strength within each corpus. All use the same zero-shot Qwen3-4B disambiguator and the same prompt. R@1 is the retriever’s top-1 accuracy, R@10 the share of mentions whose gold concept appears anywhere in the candidate list (the ceiling for the LLM stage), P4 the final Accuracy@1, and $\Delta = P4 - R@1$.

1 the LLM changes), the *intervention precision* fixes/(fixes + breaks), and Δ . Retriever confidence is the top1 – top2 score gap after P3. Score scales differ across retrievers, so confidence is always binned *within* a run, either into fixed gap bands or by percentile. Confidence intervals are 95% document-level bootstrap (Efron, 1979) resampling PMIDs, since mentions inside one abstract are correlated.

Reproducibility. Every number in this paper is recomputed from per-mention prediction dumps that record, for each mention, the retriever’s top-1 and its score gap, the pipeline’s output, whether either was correct, whether the gold concept appeared in the candidate list and at what rank. The dumps, the analysis scripts and the figure scripts are released; no model inference is needed to reproduce any table or figure here.

4 The Frame: the Retriever Sets the LLM’s Value

The same model with the same prompt contributes between +0.6 and +10.1 points depending only on what precedes it (Table 1): a seventeen-fold range with the model, the prompt and the candidate-list size held constant. The interesting question is whether that spread is systematic.

Pooled across all fourteen runs it is not clearly so: Δ correlates with Candidate Recall@1 at $\rho = -0.50$ ($p = 0.069$), because corpus difficulty

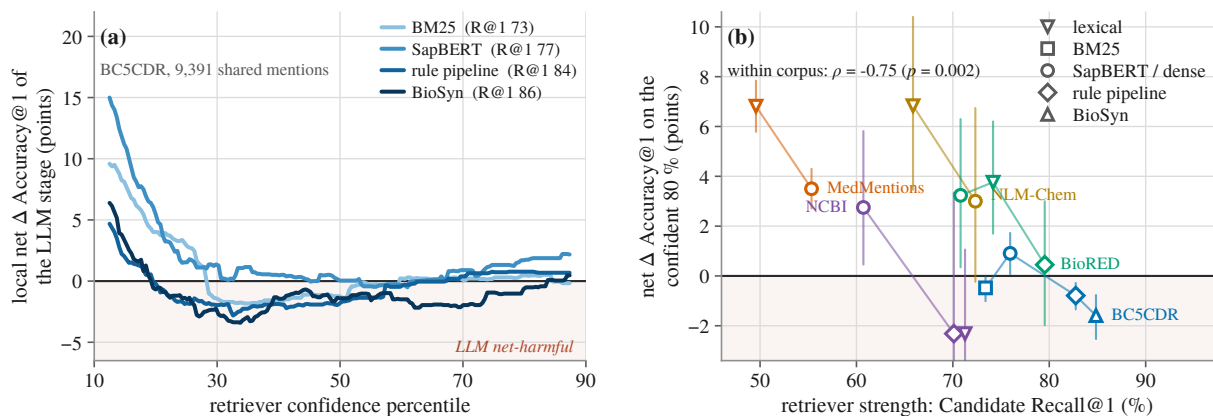


Figure 1: **Where along the retriever’s own confidence the LLM stage pays off.** (a) Local net Δ Accuracy@1 in a sliding window covering 25% of each run, over the percentile of that run’s top1–top2 score gap; BC5CDR, four retrievers, restricted to the 9,391 mentions all four cover. Every profile is strongly positive where the retriever is unsure, and the whole curve moves down as the retriever strengthens: behind BioSyn and the rule pipeline it crosses into the net-harmful region in the *middle* of the confidence range. (b) All fourteen runs: net Δ Accuracy@1 on the 80% of mentions the retriever is most confident about, with document-level bootstrap 95% CIs. Thin lines join runs on the same corpus, which is where the effect is identified; pooled across corpora the relation is confounded by how hard each corpus is.

confounds the comparison. The effect is identified *within* a corpus, where the mention set and the knowledge base are held fixed: there, 12 of the 14 ordered pairs put the smaller gain behind the stronger retriever (sign test $p = 0.013$), and after removing corpus means the relation is $\rho = -0.72$ ($p = 0.004$; OLS slope -0.32 gain points per point of Recall@1, $r = -0.76$). The two exceptions are cases where the “stronger” retriever also changed the candidate pool substantially. De-duplicating the test sets keeps the correlation ($\rho = -0.67$, $p = 0.009$) but not the sign test (10/14, $p = 0.18$), so we state the relation as graded rather than as a rule that holds pair by pair — and in §4.1 we stop inferring it and manipulate it.

A direct consequence is that the two components are partial *substitutes*: on the same MedMentions mentions, moving from lexical to SapBERT retrieval buys +7.3 points of Recall@1 but only +2.7 points of final accuracy, and on BC5CDR replacing SapBERT with BioSyn buys +8.9 and +4.9. Investment in the retriever has diminishing final returns because the LLM stage was masking the retriever’s weakness. §4.1 measures that substitution directly.

Figure 1a shows where in the pipeline this happens. With the retriever varied on a fixed BC5CDR mention set, every configuration’s value profile is strongly positive at low retriever confidence and flat-to-negative at high confidence; what changes with retriever strength is the *level* of the whole

curve. Summarising each run by its Δ on the confident 80% of its mentions (Figure 1b) gives the same within-corpus relation, more sharply: $\rho = -0.75$ ($p = 0.002$; $r = -0.85$, $p = 0.0001$). Behind the strongest retrievers this quantity is negative — the stage is a net cost on everything except the mentions the retriever is least sure about.

4.1 A controlled dose-response

The relation above is observational: swapping retrievers moves the candidate pool, the score scale and top-1 accuracy together. To manipulate one factor alone we take BioSyn’s BC5CDR candidate lists and, for a fraction p of mentions, swap the concept at rank 1 with a uniformly drawn rank in $[2, 10]$, leaving the score vector in place. The candidate set, the score distribution and the confidence signal are therefore identical at every level — Recall@1@10 is 90.5 throughout — and only the retriever’s top-1 accuracy falls, from 84.8 at $p = 0$ to 42.4 at $p = 0.5$.

Two things follow (Figure 2). First, the stage absorbs the damage: a 42.4-point loss of Recall@1 costs only 4.8 points of final accuracy, so the LLM recovers 89% of what the retriever gives up, and its gain is an almost exactly linear function of retriever quality (slope -0.89 points per point, $r = -1.0000$). That is the substitution above, measured causally rather than inferred across corpora. Second, the effect on the confident majority is linear in the same way and *changes sign*:

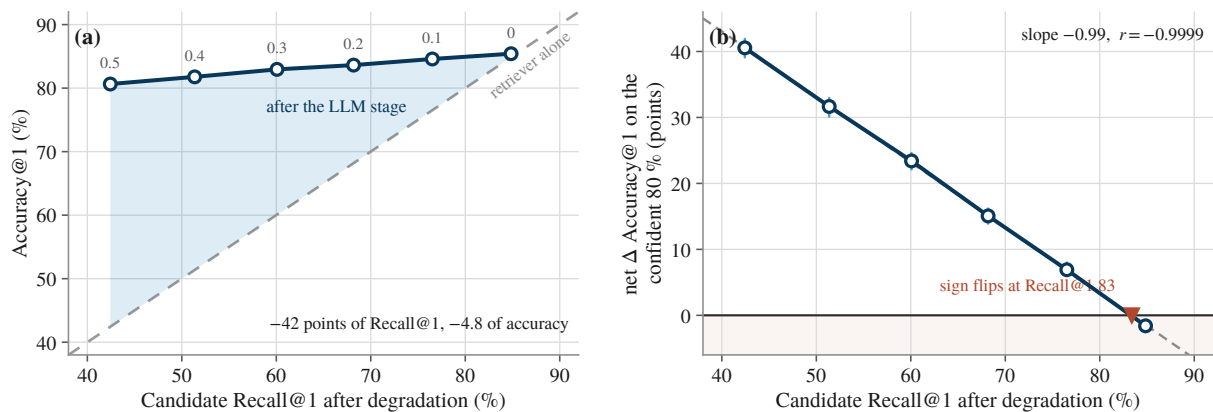


Figure 2: **Graded retriever degradation on a fixed candidate pool.** Starting from BioSyn’s BC5CDR candidate lists, the concept at rank 1 is swapped with a uniformly drawn rank in $[2, 10]$ for a fraction p of mentions. The score vector stays in place, so the list is still sorted, the top1–top2 gap distribution is unchanged, and Recall@1@10 is 90.5 at every level *by construction*. Only which concept sits on top changes. (a) The retriever loses 42.4 points of Recall@1; final accuracy loses 4.8. The shaded wedge is the LLM stage’s contribution, and it grows exactly as fast as the retriever’s loss. (b) On the confident 80% of mentions the stage’s effect is a linear function of retriever quality that crosses zero at Recall@1 $\approx$ 83.

it runs from +40.6 points at $p = 0.5$ to -1.6 at $p = 0$, crossing zero at Recall@1 $\approx$ 83 (slope -0.99 , $r = -0.9999$). The danger zone is not a property of a particular retriever but of retriever accuracy itself.

One thing this design deliberately does not vary is recall: degrading the ranking leaves the gold concept in the list, whereas a genuinely weaker retriever also retrieves it less often (BM25 reaches Recall@1@10 79.1 against BioSyn’s 90.5). The controlled slope of -0.89 is therefore an upper bound on what swapping real retrievers buys, and the smaller observational slope of -0.32 is what one should expect when recall and ranking move together.

5 A Mid-Confidence Danger Zone

5.1 The mechanism

The zero-shot model’s intervention *rate* behaves sensibly: on BC5CDR it falls monotonically from 54.3% of the most-uncertain mentions to 0.9% of the most-certain ones, and likewise on every other corpus (Figure 3, line). It acts less when the retriever is sure.

Its *precision* does not track its rate. It is near-perfect where the retriever is unsure (BC5CDR 98% at gap $[0, 2)$) but collapses in a mid-high band rather than at the extreme: at gap $[10, 20)$ the model still intervenes on 6.1% of mentions and is right only 18% of the time (95% CI $[10, 30]$; 20 fixes against 89 breaks), turning a stage that is strongly

positive elsewhere into a net -2.4 points there. The retriever is already correct on 86.5% of those mentions, so most interventions overturn a correct answer. The naive reading — “the more confident the retriever, the more harmful the LLM” — is wrong: the topmost band (gap 20+) is *not* net-harmful (+0.2), and its precision is too noisy to interpret (rate $< 1\%$, CI $[14, 84]$). The harm is localised to a middle band where the model is confident enough to act while the retriever is already reliable.

5.2 Where it replicates and where it does not

The net-harmful band reproduces on BioRED (-1.4 ; precision 39%, CI $[13, 76]$), NCBI-Disease (-7.6 ; every one of the 29 interventions in that band is a break or lateral, 0 fixes against 19 breaks, on 249 mentions) and NLM-Chem (-1.0 ; precision 38%, CI $[13, 75]$), but not equally robustly: under the de-duplicated protocols of Appendix A the band survives on BC5CDR and NCBI-Disease and disappears on BioRED and NLM-Chem, where it rests on a few hundred mentions. We therefore claim replication for BC5CDR (two retrievers) and NCBI-Disease, and report BioRED and NLM-Chem as protocol-dependent. On MedMentions, linked against the full UMLS, no band is net-harmful under any protocol.

For NLM-Chem we can be more definite than “protocol-dependent”, because the earlier figure came from a 1,500-mention subsample. Re-running the zero-shot baseline over the full test set puts 3,623 mentions in the band — a $7.6\times$ in-

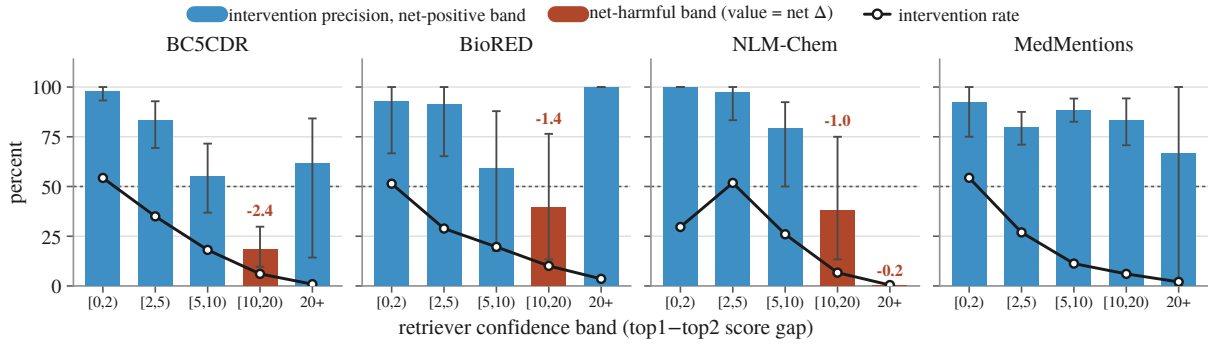


Figure 3: **The intervention rate is calibrated; the intervention precision is not.** Per confidence band: how often the zero-shot LLM changes the retriever’s top-1 (line) and how often such a change is an improvement (bars, document-level bootstrap 95% CI). The rate falls monotonically, but precision collapses in a mid-high band, not at the extreme; bands where the stage is net-harmful are marked with their Δ . The pattern is absent on MedMentions, where the retriever is too weak for an override to cost anything.

crease — and the band is still not distinguishable from zero: $\Delta = -0.33$, document-level bootstrap CI $[-2.68, +1.69]$. Notably the interval barely narrows (it was $[-3.96, +1.44]$ on the subsample), because NLM-Chem’s 50 full-text documents, not its mentions, are the resampling unit and therefore the binding constraint. The absence of a band on NLM-Chem is thus a result rather than a sample-size artefact, and no larger run on this corpus would settle it further. This is what the mechanism predicts: an inversion only costs accuracy when the retriever’s mid-confidence answers are usually right. On MedMentions the retriever is weak (Recall@1 49.6–55.3%), so even its moderately confident top-1 is often wrong and the LLM has room to help; on the MeSH corpora the mid-confidence top-1 is mostly right, so overriding it is expensive.

To check that this is not an artefact of our own rule-based confidence score, we repeat the BC5CDR analysis with candidates and scores taken purely from two public retrievers. On the weaker *SapBERT* bi-encoder (Recall@1 75.9%, the retriever used by Ye and Mitchell, 2025 and Lin et al., 2025) every band is net-positive. On the stronger *BioSyn* (Recall@1 84.8%) the danger zone reappears with an entirely external confidence signal: precision falls to 34% and 23% in the $[5, 10)$ and $[10, 20)$ bands, net -2.5 and -1.7 points, while the low-confidence bands stay strongly positive ($+17.7$, $+7.2$). The effect therefore tracks retriever strength rather than our implementation. It is an instance of what Doku (2026) call an inversion zone — a region where higher confidence does not mean the intervention is safer — although we do not rely on that work’s formal apparatus for any claim here.

Not a surface-form artefact. One might suspect the score gap is a proxy for how easy the mention string is, so that the danger zone only says the LLM should not touch exact string matches. The gap correlates with mention-label surface similarity only moderately ($\rho = 0.38$ – 0.50), and the band structure survives stratifying on it: among the BC5CDR mentions below surface similarity 80 — the hard forms, where the stage does essentially all of its work — Δ runs $+22.8$, $+13.1$, $+1.2$, -4.6 , $+0.9$ across the five bands, with precision 99, 85, 54, 18, 64%. Above similarity 95 the model intervenes on 0.2% of mentions, so nothing happens there either way.

5.3 Robustness: the prompt

The obvious objection to a behavioural finding is that the prompt invited the behaviour. It did not. The prompt used for every run in this paper already contains the instruction “*the top-ranked candidate is correct in roughly 80% of cases; override it only when the context gives you clear evidence*”, and it already lists each candidate’s retriever score, from which the margin is one subtraction away. The danger zone appears anyway.

To measure what those two ingredients do, we re-run BC5CDR and BioRED with three single-factor variants (Figure 4). **Removing the scores** (v8-noscore) is clearly harmful: the intervention rate rises ($8.0 \rightarrow 11.7\%$ on BC5CDR, $13.0 \rightarrow 18.2\%$ on BioRED), precision falls ($61 \rightarrow 56\%$, $75 \rightarrow 63\%$), the net-harmful band deepens ($-2.4 \rightarrow -3.2$, $-1.4 \rightarrow -6.0$) and BioRED loses 1.1 points of accuracy ($p = 0.041$). The numeric confidence signal is load-bearing. **Removing the verbal in-**

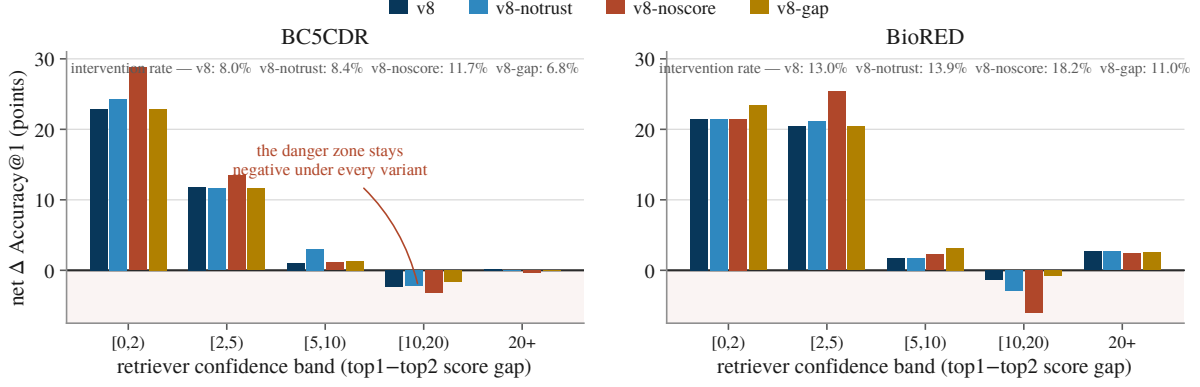


Figure 4: **The danger zone survives every prompt intervention we tried.** Net Δ Accuracy@1 per confidence band for the prompt used throughout the paper (v8) and three single-factor variants. v8 already tells the model to trust the ranking *and* shows it every candidate’s match score. Removing the scores makes the stage act more and decide worse; removing the verbal instruction changes almost nothing; stating the margin as a number and telling the model how to use it halves the net-harmful band without removing it.

struction (v8-notrust) does almost nothing: +0.24 points on BC5CDR ($p = 0.054$) and -0.42 on BioRED ($p = 0.21$), neither significant. **Stating the margin explicitly** as a number, with a rule for how to use it (v8-gap), moves the model in the right direction — the intervention rate falls to 6.8% and 11.0%, precision rises to 65% and 80% — and it halves the net-harmful band ($-2.4 \rightarrow -1.7$, $-1.4 \rightarrow -0.8$) but does not remove it, and does not significantly change accuracy.

The reading we draw is narrow and, we think, the interesting one: the model responds to a *quantitative* confidence signal and barely at all to a verbal one, and even when the margin is handed to it as a number together with an instruction to defer on large margins, it still over-rides a mostly-correct retriever often enough to lose accuracy. The inversion is not a prompt-engineering oversight.

Sensitivity to the evaluation protocol. Scoring every annotated mention occurrence over-weights frequent mention strings (Tutubalina et al., 2020), which changes *which* mentions dominate each band. We therefore recompute everything under two stricter protocols — unique (document, mention, concept) triples, and globally unique mention–concept pairs (Appendix A). The band is unchanged on BC5CDR, where on BioSyn it in fact deepens under de-duplication ($-2.5 \rightarrow -5.0$ at gap [5, 10)), and unchanged on NCBI-Disease; it vanishes on BioRED and NLM-Chem, whose [10, 20) bands hold only 205 and 174 unique pairs. The deference result of §6 is robust and in fact cleaner on unique pairs.

Corpus	Disambiguator	gain	band	agg.	skill
BC5CDR	Qwen3-4B	+0.9	-2.4	0.45	1.36
	Qwen3-8B	+1.8	-1.2	0.46	2.14
	Granite-8B	-1.5	-4.1	1.05	2.07
BioRED	Qwen3-4B	+4.0	-1.4	0.80	3.13
	Qwen3-8B	+4.6	-1.4	0.71	2.94
	Granite-8B	+1.8	-3.5	1.12	2.35
NCBI	Qwen3-4B	+2.4	-7.6	1.12	0.00
	Qwen3-8B	+6.5	-3.2	0.81	1.60
	Granite-8B	+2.2	-9.2	1.81	1.45
NLM-Chem	Qwen3-4B	+5.1	-0.3	0.50	3.94
	Qwen3-8B	+6.9	+0.8	0.45	6.25
	Granite-8B	+5.2	-0.7	0.90	3.86

Table 2: Three disambiguators on identical mentions, candidate lists, prompt and decoder. “gain” is Δ Accuracy@1 over the whole corpus, “band” is Δ in gap [10, 20); both are **bold** when a paired document-level bootstrap CI excludes zero. “agg.” is the intervention rate in that band divided by the retriever’s error rate there, and “skill” is intervention precision divided by the same quantity; neither uses p4, so neither restates the band column. Agg. above 1 (bold) means the model intervenes more often than the retriever is wrong and must therefore over-ride correct predictions.

5.4 Robustness: the disambiguator

The results so far rest on one 4B model. We repeat the core measurement with **Qwen3-8B** (same family, double the size) and **Granite-3.1-8B-Instruct** (different family, same size as Qwen3-8B) on four corpora, holding the mentions, candidate lists, prompt, temperature and constrained decoder fixed (Table 2).

Across the twelve model–corpus cells the net-harmful band is significantly negative in five, indistinguishable from zero in seven, and sig-

nificantly positive in none. It is not a property of Qwen3-4B: Granite, a different family at twice the size, shows it more strongly than our base model does (NCBI-Disease -9.2 , CI $[-14.1, -4.7]$; BC5CDR -4.1 , CI $[-5.8, -2.6]$). Nor is it a property of small models. Qwen3-8B does attenuate it significantly relative to the 4B (BC5CDR $+1.18$, CI $[+0.23, +2.20]$; NCBI-Disease $+4.42$, CI $[+0.79, +9.41]$), but Granite at the same parameter count is worse than the 4B everywhere. Scale is not the variable; the model is.

One cell deserves emphasis because it is the frame of §4 realised without any manipulation on our part. **On BC5CDR, Granite’s disambiguation stage makes the pipeline significantly worse:** accuracy 81.3 against a retriever at 82.7, a net -1.5 with CI $[-2.6, -0.2]$. Behind the strongest retriever in our set, a competent, widely used 8B instruction model is a net cost.

The two right-hand columns say why, and they are the sharper version of §5’s mechanism. Normalising against the retriever’s error rate in the same band separates *how much* a model intervenes from *how well* it chooses, and neither quantity uses the outcome. The Qwen models modulate: on BC5CDR the aggressiveness ratio falls $0.66 \rightarrow 0.53 \rightarrow 0.52 \rightarrow 0.45 \rightarrow 0.15$ across the five bands, so they act far less often, relative to the opportunity, where the retriever is reliable. Granite does not: its profile is 0.91, 0.85, 0.86, 1.05, 0.93 — essentially flat, and above 1 in the harmful band, where intervening more often than the retriever errs *guarantees* over-riding correct answers. The same holds on BioRED and NCBI-Disease. Ranked by harm, the four worst cells in Table 2 (-9.2 , -7.6 , -4.1 , -3.5) are exactly the four with an aggressiveness ratio above 1.

Granite is not worse at *choosing*. Its selection skill in the harmful band (2.07 on BC5CDR) is above the 4B’s (1.36); both are far below the ratio needed to break even there (3.7). What separates the models is not judgement but allocation — the same distinction fine-tuning turns out to act on (§6).

5.5 The consequence: a confidence gate

Because the harm is confined to a predictable band, a simple policy addresses it: call the LLM only when the retriever’s gap is below a threshold, otherwise keep the retriever’s answer. To avoid tuning on the reported set we split documents in half, choose the threshold on one half and report on the other, repeating over 20 random splits (Table 3).

Corpus	Retriever	full	gate	calls	gate—full
BC5CDR	BioSyn	85.1	86.3	18	$+1.21 \pm .27$
	pipeline	83.6	84.2	14	$+0.55 \pm .27$
	SapBERT	80.4	80.4	99	$+0.01 \pm .01$
BioRED	pipeline	83.9	82.8	47	$-1.11 \pm .92$
NCBI	pipeline	72.0	73.4	31	$+1.43 \pm 1.5$
NLM-Chem	dense	77.7	78.2	41	$+0.54 \pm .42$
MedMent.	SapBERT	59.7	59.6	92	$-0.03 \pm .04$
<i>fine-tuned control</i>					
BC5CDR	pipeline	86.2	86.1	79	$-0.14 \pm .17$

Table 3: Confidence gate: call the LLM only when the retriever’s top1—top2 gap is below a threshold tuned on half the *documents* and evaluated on the other half. “full” is the accuracy of calling the LLM on every mention, “gate” that of the gated system and “calls” the share of mentions it sends to the LLM; all are means over 20 random document splits, and the last column carries the standard deviation of the difference across splits. The gate helps where the retriever is strong enough to have produced a danger zone, is neutral where it is not, and is harmful on BioRED. On the fine-tuned model it no longer helps, because fine-tuning has removed the band it exploits (§6).

The result is two-sided, and we state the negative half plainly. Behind the two strongest BC5CDR retrievers the gate *beats* calling the LLM everywhere: $+1.21 \pm 0.27$ points at 18% of the calls on BioSyn and $+0.55 \pm 0.27$ at 14% on the rule pipeline. Behind the weaker SapBERT retriever it is exactly neutral and selects essentially every mention (99% of calls), as does MedMentions (92%) — there is no band to remove. On BioRED the gate is *worse* than the full LLM by 1.11 ± 0.92 points: the corpus has a net-harmful band, but it is small relative to the gains the stage makes elsewhere, and a single scalar threshold cannot excise it without also discarding those. The honest summary is not “less is more” but “gating pays exactly where a strong retriever has created an inversion zone, and should be validated per deployment rather than assumed”.

6 Fine-Tuning Buys Deference, Not Competence

Fine-tuning raises accuracy on all four corpora (Table 4). The question is *what* it improves. Every number in this section is the mean over three training seeds, which differ only in weight initialisation and data order; the ranges matter, because a single run overstates two of the effects below.

Competence does not improve; allocation does. Read by confidence band (Figure 5a), fine-tuning

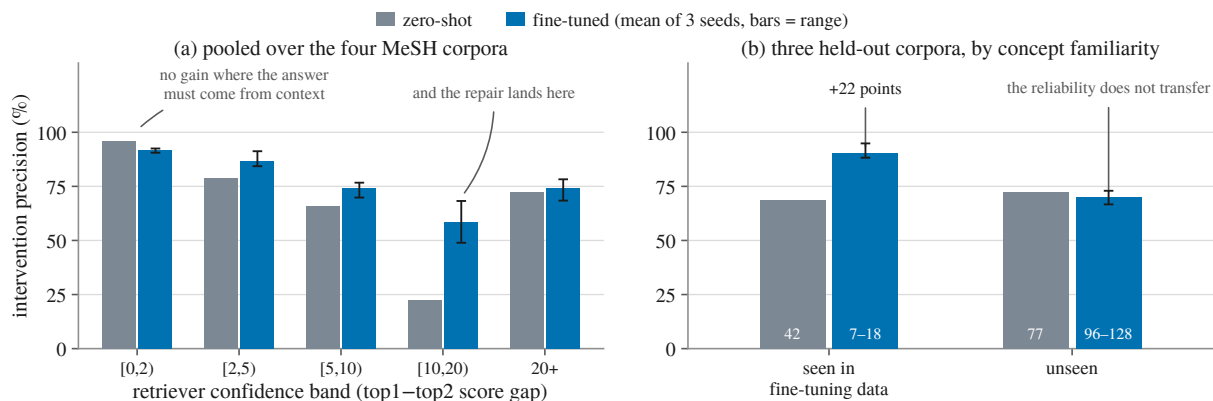


Figure 5: **What LoRA fine-tuning changes.** Means over three training seeds; error bars give the range. **(a)** Intervention precision per confidence band, pooled over the four MeSH corpora. On the truly-uncertain band the fine-tuned model is no more precise than the zero-shot one — slightly less so; the gains sit in the bands where the retriever is already mostly right. **(b)** Intervention precision on three corpora the model never trained on, split by whether the gold concept occurs in the fine-tuning data. The reliability fine-tuning buys is realised on seen concepts and is absent on unseen ones, where precision in fact declines; numbers inside the bars are breaks.

Corpus	R@1	P4		prec.		<i>p</i>
		zs	ft	zs	ft	
BC5CDR [†]	82.7	83.7	86.5 ^{86.8} _{86.0}	61	82 ⁸⁷ ₇₇	<.001
BioRED	79.5	83.6	85.0 ^{85.1} _{84.8}	75	77 ⁸⁰ ₇₅	≤.028
NCBI	70.1	72.5	77.9 ^{79.4} _{76.9}	58	74 ⁸² ₇₀	<.001
NLM-Chem	72.3	78.0	79.4 ^{79.9} _{78.6}	80	76 ⁷⁸ ₇₃	.005–.43

Table 4: Zero-shot (zs) vs. LoRA fine-tuned (ft) disambiguator on the identical mentions and identical candidate lists; “prec.” is overall intervention precision. Fine-tuned entries are the mean over **three training seeds**, with the observed range over seeds set as the sub/superscript, and *p* is the largest (worst) exact McNemar test over those three seeds. [†]in-domain: the model was fine-tuned on the BC5CDR training split; the other three corpora are held out. Accuracy improves on every corpus and every seed, but on NLM-Chem the improvement is not significant on one of the three (*p* = .43), and overall intervention precision improves only where it was low to begin with — on BioRED and NLM-Chem, where the zero-shot model was already precise, fine-tuning does not add to it and on NLM-Chem slightly reduces it.

does not make the model a better disambiguator where disambiguation is hard. On the truly-uncertain band [0, 2), where the answer has to come from context, pooled intervention precision does not rise — it falls slightly, 96.1% → 91.6% (range 90.6–92.5). The model acts *more* often there and nets more repairs (fixes 149 → 228), but its accuracy per intervention is no better than before, and breaks appear where the zero-shot model had almost none (6 → 21).

What changes is where the interventions go. Fine-tuning does not make the model quieter: on BC5CDR the overall intervention rate *rises*, 8.0% → 10.0%. It redirects them. In the band [10, 20), where the retriever is already right on 86.5% of mentions, precision goes from 18% to 62% (range 50–74) and the net effect from $\Delta = -2.4$ to $\Delta = +0.7$ (range -0.0 to $+1.3$); in the two bands where the retriever is usually wrong the model intervenes markedly more. Fine-tuning buys the gate of §5, learned internally, and consistently the external gate no longer beats the full LLM on the fine-tuned model (Table 3, last row).

The band repair is not universal, and here the seeds correct a single-run result. It is strong on BC5CDR and on NLM-Chem (38% → 72%, range 67–76; $\Delta -1.0 \rightarrow +2.9$), partial and still centred on zero on NCBI-Disease (0% → 50%, range 36–70; $\Delta -7.6 \rightarrow -0.4$, range -1.6 to $+1.6$), and *absent* on BioRED, where precision in that band stays at 39% (range 38–41) and the band remains net-harmful ($-1.4 \rightarrow -1.2$). A single fine-tuning run had suggested 69% on BioRED; none of the three seeds reproduces it. BioRED is the same corpus on which the danger zone did not survive de-duplication (§5) and on which the confidence gate loses (Table 3) — three independent checks pointing the same way.

The restraint is bound to the training distribution. Whether this restraint *transfers* depends on whether the concept was in the fine-tuning data. Pooling the three corpora the model never trained

on (BioRED, NCBI-Disease, NLM-Chem; 4,024 mentions), fine-tuning raises intervention precision on *seen* concepts from 68.7% to 90.5% (range 88.2–94.9 over the three seeds), while on *unseen* concepts it does not improve at all: 72.2% → 69.8% (range 66.7–73.0). The two seed ranges do not overlap, and the direction differs: breaks on seen concepts fall from 42 to 7–17, while breaks on unseen concepts *rise* from 77 to 96–128.

The band breakdown localises it. On seen concepts the gain is confined to exactly the bands where the danger zone lived — +23 points at gap [5, 10) and +50 at [10, 20), with [0, 2), [2, 5) and 20+ unchanged. On unseen concepts no band gains: −11, +7, −4, +5, −27 across the five. Unlike the danger zone, this contrast does not depend on the evaluation protocol; on globally unique mention–concept pairs it is sharper still, 65.9% → 87.5% (range 86.2–89.5) on seen concepts against 70.6% → 64.2% (range 60.9–66.7) on unseen ones, where the unseen side moves clearly *down*.

We therefore make the claim the data support: fine-tuning installs a confidence-dependent restraint, and that restraint is a property of the concepts it was trained on rather than a policy it applies to new ones. Accuracy still improves out-of-domain (Table 4), so the model is not worse overall on novel concepts — but it buys that improvement by intervening more, not by choosing better, and the near-perfect reliability it appears to acquire does not come with it.

7 An Anatomy of the Interventions

The zero-shot model changes the retriever’s top-1 on 8.0% of BC5CDR mentions (771 of 9,661): 33% fixes, 21% breaks, 46% lateral. Splitting the lateral moves matters, because only 14.5 of those 46 points occur when the gold concept was in the candidate list at all — genuine confusion — while the remaining 31.5 are lateral by construction, the gold having never been retrieved. The genuine-confusion rate is therefore about 15% of all changes, not 46% (Appendix B). Two quantities bound what the stage can do: of the retriever’s errors whose gold *is* in the candidate list it recovers 24% (MedMentions) to 66% (BC5CDR/BioSyn), 30% on the BC5CDR rule pipeline; and on that run 809 of the 1,667 retriever errors have no gold concept in the list, so no disambiguator could fix them. Averaged over the three fine-tuning

seeds the picture sharpens the account of §6: the model makes *more* changes (771 → 970), raises fixes to 48% (range 44–50) and halves breaks to 10% (range 8–13), while genuine confusion barely moves (14.5% → 9.9%, range 9.7–10.2) and the share of changes that were lost before the LLM saw them is unchanged at 32%. Recovery on the repairable errors rises from 29% to 54% (range 53–56) — almost entirely by attempting more of them, not by confusing fewer. The stage is best described as over-eager where there is nothing to gain and under-powered where it matters. The pattern is not driven by entity type: on BC5CDR the stage adds +1.0 points on the 5,351 chemical mentions and +0.9 on the 4,310 disease mentions.

8 Discussion

One account ties the results together. The LLM disambiguation stage is not a general semantic reasoner applied uniformly to a candidate list; it is a retriever-dependent, mechanically bounded correction, valuable exactly where the retriever is uncertain and idle-to-harmful on the confident majority. Fine-tuning does not lift its ceiling on the hard cases; it teaches the model *when* to defer, and that deference is calibrated to the concepts it was trained on.

Three consequences follow for practice. Reported gains from an LLM disambiguation stage are statements about a retriever–LLM pair and should be reproduced against the retriever the authors used before being attributed to the model. Behind a strong retriever the stage should be gated or cascaded, but the gate must be validated on the target corpus — it helped on three of our seven configurations, was neutral on two and hurt on one. And a fine-tuned disambiguator should be expected to be reliable on familiar concepts and merely adequate on novel ones, which is precisely the regime biomedical deployment cares about.

Limitations

No non-LLM re-ranker baseline. The zero-shot results cover three disambiguators from two families (§5.4) and the fine-tuning results three seeds (§6), so the effects are not properties of one model or one run. What we do not have is a *non-generative* comparison: a cross-encoder re-ranker over the identical candidate lists. Cross-encoders are the established strong re-ranker for biomedical EL, and without one we cannot say whether

the behaviour we describe is specific to LLM disambiguators or general to any learned re-ranking stage placed behind a strong retriever. We regard this as the most important open comparison, and note that either outcome would sharpen rather than undermine the account: a cross-encoder without an inversion zone would make the practical recommendation concrete, and one with an inversion zone would widen the claim beyond LLMs.

Fine-tuning is one recipe. The deference result uses LoRA SFT on the gold answer, over three seeds. It does not tell us what a different objective would buy. The distinction the paper keeps arriving at — that the failure is allocation rather than judgement — suggests a reward that scores the *decision* (repair, break, abstain) rather than the answer, which supervised fine-tuning on gold labels cannot express. We have not tested that.

The degradation varies ranking, not recall. The dose-response of §4.1 holds Recall@1@10 fixed by construction, so it measures the stage’s ability to repair a mis-ranked candidate list, not its value behind a retriever that fails to retrieve the gold concept at all. Real retrievers differ on both axes, which is why the controlled slope (−0.89) exceeds the observational one (−0.32). A second manipulation that removes candidates rather than reordering them would complete the picture; we have not run it.

The seen/unseen split is confounded with frequency. Concepts present in the BC5CDR training data are also the more frequent ones, so part of the seen-concept improvement in §6 may reflect frequency rather than exposure. A clean isolation stratifies within frequency bands; we have not run it.

Evaluation protocol. Following BioLinkerAI we score every annotated mention occurrence, which inflates absolute accuracies relative to de-duplicated protocols (Tutubalina et al., 2020) and makes our absolute numbers not directly comparable to published leaderboards. Appendix A repeats the analysis under two stricter protocols and names the two claims that do not survive them. Separately, the BM25 run covers 9,391 of the 9,661 BC5CDR mentions; where the four BC5CDR retrievers are compared directly (Figure 1a) we restrict to the shared subset.

Confidence signal. Our confidence signal is a top1–top2 score gap, which is neither calibrated nor comparable across retrievers; we handle this by binning within each run, but a calibrated retriever confidence would be a stronger instrument and might move the location of the danger zone.

Scope. We evaluate linking given gold mentions, on English abstracts and one full-text corpus, against MeSH and UMLS. Nothing here speaks to mention detection, to other knowledge bases, or to languages other than English.

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References

- Olivier Bodenreider. 2004. [The Unified Medical Language System \(UMLS\): integrating biomedical terminology](#). *Nucleic Acids Research*, 32(suppl_1):D267–D270.
- Lingjiao Chen, Matei Zaharia, and James Zou. 2023. [FrugalGPT: How to use large language models while reducing cost and improving performance](#). *arXiv preprint arXiv:2305.05176*.
- Rezarta Islamaj Doğan, Robert Leaman, and Zhiyong Lu. 2014. [NCBI disease corpus: A resource for disease name recognition and concept normalization](#). *Journal of Biomedical Informatics*, 47:1–10.
- Ronald Doku. 2026. [The confidence gate theorem: When should ranked decision systems abstain?](#) *arXiv preprint arXiv:2603.09947*.
- Bradley Efron. 1979. [Bootstrap methods: Another look at the jackknife](#). *The Annals of Statistics*, 7(1):1–26.
- Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. *Advances in Neural Information Processing Systems (NeurIPS)*.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*.
- Rezarta Islamaj, Robert Leaman, Sun Kim, Dongseop Kwon, Chih-Hsuan Wei, Donald C. Comeau, Yifan Peng, David Cissel, Cathleen Coss, Carol Fisher, Rob Guzman, Preeti Gokal Kochar, Stella Koppel, Dorothy Trinh, Keiko Sekiya, Janice Ward, Deborah Whitman, Susan Schmidt, and Zhiyong Lu. 2021. [NLM-chem, a new resource for chemical entity recognition in PubMed full text literature](#). *Scientific Data*, 8:91.

Chanhwi Kim, Hyunjae Kim, Sihyeon Park, Jiwoo Lee, Mujeen Sung, and Jaewoo Kang. 2025. Learning from negative samples in biomedical generative entity linking. In *Findings of the Association for Computational Linguistics: ACL 2025*. Association for Computational Linguistics.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. 2023. [Efficient memory management for large language model serving with PagedAttention](#). In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*.

Jiao Li, Yueping Sun, Robin J. Johnson, Daniela Sciaky, Chih-Hsuan Wei, Robert Leaman, Allan Peter Davis, Carolyn J. Mattingly, Thomas C. Wieggers, and Zhiyong Lu. 2016. [BioCreative V CDR task corpus: a resource for chemical disease relation extraction](#). *Database*, 2016:baw068.

Zhenxi Lin, Ziheng Zhang, Jian Wu, Yefeng Zheng, and Xian Wu. 2025. Guiding large language models for biomedical entity linking via restrictive and contrastive decoding. In *Findings of the Association for Computational Linguistics: EMNLP 2025*. Association for Computational Linguistics.

Fangyu Liu, Ehsan Shareghi, Zaiqiao Meng, Marco Basaldella, and Nigel Collier. 2021. [Self-alignment pretraining for biomedical entity representations](#). In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4228–4238. Association for Computational Linguistics.

Lajanugen Logeswaran, Ming-Wei Chang, Kenton Lee, Kristina Toutanova, Jacob Devlin, and Honglak Lee. 2019. [Zero-shot entity linking by reading entity descriptions](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 3449–3460. Association for Computational Linguistics.

Ling Luo, Po-Ting Lai, Chih-Hsuan Wei, Cecilia N. Arighi, and Zhiyong Lu. 2022. [BioRED: a rich biomedical relation extraction dataset](#). *Briefings in Bioinformatics*, 23(5):bbac282.

Sunil Mohan and Donghui Li. 2019. [MedMentions: A large biomedical corpus annotated with UMLS concepts](#). In *Proceedings of the 2019 Conference on Automated Knowledge Base Construction (AKBC)*.

Ahmad Sakor, Kuldeep Singh, and Maria-Esther Vidal. 2025. [BioLinkerAI: Capturing knowledge using LLMs to enhance biomedical entity linking](#). In *Web Information Systems Engineering – WISE 2024*, volume 15439 of *Lecture Notes in Computer Science*, pages 262–272, Singapore. Springer.

Mujeen Sung, Hwisang Jeon, Jinhyuk Lee, and Jaewoo Kang. 2020. [Biomedical entity representations with synonym marginalization](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational*

Run	Band	net Δ (n)		
		occ.	doc.	uniq.
BC5CDR pipeline	[10, 20)	−2.4	−2.2	−1.5
BC5CDR BioSyn	[5, 10)	−2.5	−3.9	−5.0
BC5CDR BioSyn	[10, 20)	−1.7	−1.4	−1.5
NCBI	[10, 20)	−7.6	−5.3	−6.2
BioRED	[10, 20)	−1.4	+0.8	+1.0
NLM-Chem	[10, 20)	−1.0	± 0.0	± 0.0

Table 5: The net-harmful band under three evaluation protocols: every annotated mention occurrence (occ., the BioLinkerAI protocol used throughout the paper), unique (document, mention, concept) triples (doc.), and globally unique mention–concept pairs (uniq.). The band is protocol-independent on BC5CDR and NCBI-Disease and protocol-dependent on BioRED and NLM-Chem.

Linguistics, pages 3641–3650. Association for Computational Linguistics.

Elena Tutubalina, Artur Kadurin, and Zulfat Miftahutdinov. 2020. [Fair evaluation in concept normalization: a large-scale comparative analysis for BERT-based models](#). *Proceedings of the 28th International Conference on Computational Linguistics (COLING)*, pages 6710–6716.

An Yang et al. 2025. [Qwen3 technical report](#). *arXiv preprint arXiv:2505.09388*.

Christophe Ye and Cassie S. Mitchell. 2025. LLM as entity disambiguator for biomedical entity-linking. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*. Association for Computational Linguistics.

A Evaluation protocols

B Anatomy of the interventions

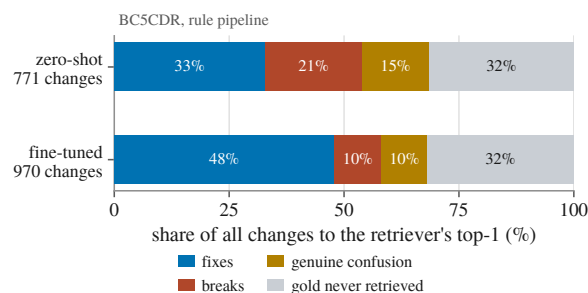


Figure 6: Every change the LLM makes to the retriever’s top-1, decomposed. Lateral moves are split by whether the gold concept was in the candidate list at all: where it was not, any change is lateral by construction and is not evidence of confusion.

C Per-band numbers

Table 6 gives the full band breakdown for the BC5CDR rule pipeline that Figure 3 summarises. Equivalent tables for the remaining thirteen runs, all prediction dumps, and the analysis and figure scripts are released with the paper.

Band	n	rate	prec.	95% CI	Δ
[0, 2)	337	54.3	98	[93,100]	+22.8
[2, 5)	543	35.0	83	[69,93]	+11.8
[5, 10)	988	18.1	55	[37,72]	+1.0
[10, 20)	2,881	6.1	18	[10,30]	-2.4
20+	4,912	0.9	62	[14,84]	+0.2

Table 6: BC5CDR, rule pipeline, zero-shot Qwen3-4B. Rate and precision in percent, Δ in Accuracy@1 points. The 20+ band’s precision rests on 34 interventions and should not be read as a recovery.